

HOPE WORLD

Software Requirements Specification

Initial Product Scope — Academic Island

Document Status	Client Review Draft
Version	1.0
Product	Hope World
Current Scope	Academic Island; Computer Island specified as a future phase
Primary Audience	Client, Product Owner, UX/UI, Engineering, QA

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1. Introduction

1.1 Purpose

This document defines the software requirements for the initial implementation of Hope World, a gamified educational platform designed primarily for children. It describes the product scope, user roles, learning model, class-based access model, content structure, quiz behavior, scoring, progress tracking, management dashboard, non-functional expectations, and identified open items.

1.2 Intended Use

This specification is intended to provide a shared baseline between the client and the delivery team. It is written to support UX/UI design, architecture, development, QA, estimation, and future change management.

1.3 Requirement Language

The term “shall” indicates a mandatory requirement for the defined scope. The term “may” indicates an optional capability or future design choice. Items explicitly marked as TBD require confirmation before they are treated as finalized behavior.

2. Product Vision and Scope

2.1 Product Vision

Hope World shall provide an engaging, child-oriented learning experience that combines structured educational content with interactive activities and gamified navigation. The learner should experience the application as an interactive world rather than a conventional learning management system.

2.2 Interactive World Concept

The primary learner interface shall represent educational areas as themed islands or worlds. The initial product concept includes domains such as Academic, Computer, Music, and other future areas. Only the Academic Island is within the detailed current scope. The Computer Island is planned for a later phase.

2.3 Current Scope

- Academic Island.
- Arabic subject.
- English subject.
- Multiple learning levels within each subject.
- Level-specific instructional models.
- Audio, video, visual, and interactive learning content.
- Drag and Drop, Matching, Typing, and Choose quiz interactions.

- Class-based student access without student accounts or authentication.
- Instructor-managed class progression.
- Class-level scoring and progress tracking.
- Manager / Head dashboard for class monitoring.

2.4 Future Scope

- Computer Island requirements will be defined in a later phase.
- Additional themed islands may be introduced after the initial scope is validated.

3. Stakeholders and User Roles

Role	Primary Responsibility
Student	Learns through the interactive world, consumes learning content, and completes quizzes. No individual account or authentication is required.
Instructor	Works with one or more classes, starts the required level for a class, and supports the class through the shared curriculum.
Manager / Head	Monitors class performance and academic progress through management-level views and dashboards.

Role note: No additional administrative role is assumed in this specification. Content-management ownership and permissions require confirmation.

4. System Concept and User Experience

4.1 Child-Oriented Experience

The learner-facing experience shall use a visually rich, friendly, interactive design suitable for children. The interface should prioritize clarity, simple navigation, visual feedback, and age-appropriate interaction patterns.

4.2 Interactive Map / World

- The learner shall be able to navigate to available educational islands from the world view.
- Educational areas shall be represented as recognizable visual locations rather than text-only menus.
- The visual style may include playful illustrations, animated elements, characters, and other child-oriented visual cues.
- The detailed visual design, colors, layout, and animation behavior shall be defined in the UI/UX specification.

4.3 Learner Avatar

- The learner interface shall support avatar selection.
- The avatar may be displayed within the learner experience and may also be used as part of educational explanations where appropriate.

- The avatar used for the learner session is not, by itself, an authenticated user identity.

4.4 Name Entry

- The learner interface may allow the child to enter a display name for the experience.
- The display name is not intended to establish a persistent authenticated student profile.

Clarification: Whether the display name is persisted for the duration of a class session or only for the current screen/session should be confirmed during UX and implementation design.

4.5 Audio and Visual Feedback

- Interactive learning experiences shall provide appropriate visual feedback.
- Audio feedback shall be available where required by the learning content.
- Audio used for Arabic and English learning content shall be clear, natural, and age-appropriate.
- The target voice experience shall be child-friendly and should avoid an unnatural or noticeably robotic delivery.
- The final voice source, recording process, and media production standards shall be defined as part of the content specification.

5. Academic Learning Model

5.1 Academic Subjects

- Arabic.
- English.

5.2 Learning Levels

Each subject shall be divided into multiple learning levels. A level is an educational stage with its own instructional model and target learning complexity. Quiz interaction types are not the definition of a level; the same quiz interaction family may be used across multiple levels.

5.3 Example Arabic Learning Progression

Learning Stage	Illustrative Learning Model
Level 1 — Letter Introduction	Introduce letters with visual presentation, pronunciation audio, and video demonstrating mouth/lip movement used for pronunciation.
Level 2 — Letter Combination	Combine letters into simple words or basic outputs, such as “أب = ب + أ”, with an appropriate visual/avatar representation.
Level 3 — Word Combination	Combine words into simple sentences, such as “أحب + أبي = أحب أبي”, with an appropriate visual/avatar representation.

Scope note: The examples above describe the intended progression pattern and are not a finalized list of all levels or all curriculum content.

5.4 Language Audio Requirement

Audio pronunciation shall be available throughout the learning levels for both Arabic and English content as applicable to the lesson. The audio shall support learning and reinforcement of vocabulary, letters, words, and other spoken educational content.

5.5 Level Composition

Each level shall consist of:

- An instructional component that implements the learning model of the level.
- A quiz/assessment component using the supported interaction types.
- The media and visual assets required by the learning model.
- Content metadata needed for correct rendering by the child-facing frontend.

6. Functional Requirements

6.1 World and Navigation

ID	Requirement	Priority	Description
FR-WLD-001	Interactive World	High	The system shall provide an interactive world/map view representing available educational islands.

ID	Requirement	Priority	Description
FR-WLD-002	Island Navigation	High	The system shall allow the learner to enter an available educational island from the world view.

ID	Requirement	Priority	Description
FR-WLD-003	Academic Island	High	The system shall provide an Academic Island as part of the initial product scope.

ID	Requirement	Priority	Description
FR-WLD-004	Future Islands	High	The architecture and content structure should allow additional islands to be introduced without redesigning the entire learning model.

6.2 Subject and Curriculum

ID	Requirement	Priority	Description
FR-ACA-001	Subject Availability	High	The Academic Island shall provide Arabic and English subjects.

ID	Requirement	Priority	Description
FR-ACA-002	Curriculum Levels	High	Each subject shall contain a predefined sequence of learning levels.

ID	Requirement	Priority	Description
FR-ACA-003	Level Progression	High	A class shall have a current active learning level within the curriculum.

ID	Requirement	Priority	Description
FR-ACA-004	Shared Class Level	High	All students participating through the same class shall receive the same active learning level.

ID	Requirement	Priority	Description
FR-ACA-005	Level-Specific Model	High	Each learning level shall define an instructional model appropriate to its learning objective.

ID	Requirement	Priority	Description
FR-ACA-006	Increasing Complexity	High	The learning model and content complexity may increase across levels according to the defined curriculum.

6.3 Learning Content

ID	Requirement	Priority	Description
FR-LRN-001	Instructional Content	High	The system shall provide instructional content before the assessment activities of a level.

ID	Requirement	Priority	Description
FR-LRN-002	Text Content	High	The content model shall support text-based educational elements where required.

ID	Requirement	Priority	Description
FR-LRN-003	Image Content	High	The content model shall support images and visual assets.

ID	Requirement	Priority	Description
FR-LRN-004	Audio Content	High	The content model shall support audio assets for pronunciation, explanation, and child-friendly learning feedback.

ID	Requirement	Priority	Description
FR-LRN-005	Video Content	High	The content model shall support video assets, including pronunciation videos demonstrating mouth/lip movement where required.

ID	Requirement	Priority	Description
FR-LRN-006	Interactive Visuals	High	The content model shall support interactive visual elements and avatar-based educational representations.

ID	Requirement	Priority	Description
FR-LRN-007	Multi-Language Content	High	The learning content model shall support Arabic and English content as separate or language-specific content variants.

6.4 Quiz Engine

ID	Requirement	Priority	Description
FR-QIZ-001	Quiz Availability	High	The system shall provide quizzes as part of a learning level.

ID	Requirement	Priority	Description
FR-QIZ-002	Drag and Drop	High	The quiz engine shall support drag-and-drop questions.

ID	Requirement	Priority	Description
FR-QIZ-003	Matching	High	The quiz engine shall support matching questions.

ID	Requirement	Priority	Description
FR-QIZ-004	Typing	High	The quiz engine shall support typing/input questions.

ID	Requirement	Priority	Description
FR-QIZ-005	Choose	High	The quiz engine shall support choose/selection questions.

ID	Requirement	Priority	Description
FR-QIZ-006	Correctness Evaluation	High	The system shall determine whether a submitted

			answer is correct or incorrect according to the configured question answer.
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ID	Requirement	Priority	Description
FR-QIZ-007	Retry	High	The system shall allow the learner to retake a quiz.

ID	Requirement	Priority	Description
FR-QIZ-008	Question Feedback	High	The system shall provide visual and/or audio feedback appropriate to the question and learning context.

ID	Requirement	Priority	Description
FR-QIZ-009	Question Scoring	High	The initial scoring model shall be based on the number of correctly answered questions.

ID	Requirement	Priority	Description
FR-QIZ-010	Question Content Independence	High	The quiz engine shall support multiple question interaction types without requiring a separate hard-coded learning flow for each level.

6.5 Class-Based Student Access

ID	Requirement	Priority	Description
FR-ACC-001	Class Entry	High	A learner shall select or enter the name of their class before accessing class-specific learning results.

ID	Requirement	Priority	Description
FR-ACC-002	Class Association	High	Quiz results generated during the learner session shall be associated with the selected class.

ID	Requirement	Priority	Description
FR-ACC-003	No Student Login	High	The student experience shall not require username/password authentication.

ID	Requirement	Priority	Description
FR-ACC-004	No Persistent Student Profile	High	The initial release shall not require a persistent individual student profile

			for learning access or reporting.
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ID	Requirement	Priority	Description
FR-ACC-005	No Individual Performance Requirement	High	The initial reporting model shall focus on class-level performance rather than individual student performance history.

6.6 Instructor and Class Progress

ID	Requirement	Priority	Description
FR-INS-001	Class Assignment	High	Each class shall have an assigned instructor.

ID	Requirement	Priority	Description
FR-INS-002	Multiple Class Teaching	High	An instructor shall be able to teach more than one class.

ID	Requirement	Priority	Description
FR-INS-003	Start Active Level	High	The instructor shall be able to start or set the active learning level for the class.

ID	Requirement	Priority	Description
FR-INS-004	Shared Progression	High	All learners accessing the same class shall be aligned to the class's active level.

ID	Requirement	Priority	Description
FR-INS-005	Class Progress Tracking	High	The system shall record the current curriculum level reached by each class.

ID	Requirement	Priority	Description
FR-INS-006	Academic Year Context	High	Class progress shall be associated with the relevant academic-year context.

7. Content Management Requirements

The educational content shall be dynamically managed rather than hard-coded into the frontend. At the same time, content shall remain constrained by a defined structure so that the frontend can render it consistently and maintain the intended child-friendly experience.

7.1 Structured Content Model

ID	Requirement	Priority	Description
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FR-CMS-001	Dynamic Content	High	The system shall store learning content as configurable data rather than requiring code changes for normal educational content updates.
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ID	Requirement	Priority	Description
FR-CMS-002	Structured Schema	High	Learning content shall conform to a defined structural schema that identifies the subject, level, instructional elements, media, and assessment content.

ID	Requirement	Priority	Description
FR-CMS-003	Frontend Rendering	High	The frontend shall render supported content types according to their configured content structure.

ID	Requirement	Priority	Description
FR-CMS-004	Media References	High	The content structure shall support references to images, audio files, video files, and other required media assets.

ID	Requirement	Priority	Description
FR-CMS-005	Quiz Configuration	High	Quiz questions shall be configurable with their question type, content, answer definition, and scoring information.

ID	Requirement	Priority	Description
FR-CMS-006	Language Variant	High	Content shall support Arabic and English variants where language-specific content is required.

7.2 Content Management Permissions

The system requires a mechanism for authorized staff to create and update learning content. The exact role responsible for content creation and maintenance is not yet confirmed. The final permission matrix shall be defined before implementation of the content-management interface.

8. Class, Instructor, and Access Management

8.1 Class Entity

A class is the primary operational grouping used by the initial product. It represents the shared learning context for a group of students and is the unit used for progress tracking and aggregate performance.

8.2 Required Class Relationships

- A class shall have one assigned instructor in the initial model.
- An instructor may be assigned to multiple classes.
- Students do not require persistent profiles; they participate through a selected class.
- A class has one active learning level at a given point in the curriculum.
- Class performance is calculated from learning activity results associated with that class.

8.3 Access Flow

1. 1. Learner opens the child-facing application.
2. 2. Learner selects or enters the class name.
3. 3. The system validates the class reference.
4. 4. The system loads the active curriculum level for that class.
5. 5. The learner accesses the available lesson and quiz activities for that level.
6. 6. Completed quiz results are associated with the selected class.

Access constraint: The system intentionally avoids student authentication and persistent student accounts in the initial scope.

9. Scoring and Progress Tracking

9.1 Initial Scoring Principle

The first release shall use a simple scoring model centered on correctness. Each question shall be evaluated as correct or incorrect. The system shall count correct answers to produce quiz-level performance.

9.2 Learner Session Score

- A completed quiz attempt shall have a score derived from its number of correct answers.
- The system may present the resulting score to the learner as a percentage or another client-approved simple representation.
- The system shall not require a persistent individual student record to calculate a current session result.

9.3 Class Score

The initial business requirement is to represent class performance as an average of learner results. The initial release shall therefore support an aggregate class score derived from results associated with that class.

Scoring evolution: A more advanced scoring model may be introduced in a later phase. The initial implementation should keep the aggregation logic intentionally simple and replaceable.

9.4 Repeated Quiz Attempts

Learners may repeat quizzes. The system shall record the resulting attempts needed to support class scoring and progress. The exact aggregation treatment of repeated attempts (for example, latest attempt, best attempt, or all attempts) is intentionally left open because the current product decision is to keep the first scoring version simple.

9.5 Progress Tracking

ID	Requirement	Priority	Description
FR-PROG-001	Active Class Level	High	The system shall maintain the current active learning level for each class.

ID	Requirement	Priority	Description
FR-PROG-002	Level Completion	High	The system shall support recording that a class has completed or reached a curriculum level.

ID	Requirement	Priority	Description
FR-PROG-003	Academic-Year Progress	High	The system shall make it possible to determine the level reached by a class during the academic year.

ID	Requirement	Priority	Description
FR-PROG-004	Shared Progress	High	Progress shall be tracked at class level rather than as separate student learning paths.

10. Manager / Head Dashboard

10.1 Purpose

The Manager / Head dashboard shall provide a concise operational view of class performance and academic progress.

10.2 Required Metrics

ID	Requirement	Priority	Description
FR-DAS-001	Class Score Metric	High	The dashboard shall display the current overall score for each class.

ID	Requirement	Priority	Description
FR-DAS-002	Current Level Metric	High	The dashboard shall display the current learning level reached by each class.

ID	Requirement	Priority	Description
FR-DAS-003	Academic-Year Progress	High	The dashboard shall allow the Manager / Head to understand the progression of each class through the academic-year curriculum.

ID	Requirement	Priority	Description
FR-DAS-004	Class Overview	High	The dashboard shall provide a class-oriented view rather than requiring individual student performance analysis.

10.3 Dashboard Presentation

- The dashboard should make class-to-class comparison easy to understand.
- Progress indicators should use clear labels and visual cues appropriate for administrative users.
- The exact chart types, filters, date controls, and export capabilities are not yet specified and shall not be assumed.

11. Non-Functional Requirements

ID	Category	Requirement
NFR-001	Usability	The child-facing application shall prioritize simple navigation, clear interaction feedback, and age-appropriate presentation.
NFR-002	Accessibility of Interaction	Interactive controls shall provide sufficiently clear visual states so children can understand selection, success, error, and completion.
NFR-003	Performance	Common navigation and lesson transitions should feel responsive enough to preserve a continuous child-friendly experience.
NFR-004	Media Playback	Audio and video content shall be delivered in a manner that supports reliable playback during lessons.

ID	Category	Requirement
NFR-005	Scalability of Content	The content model should support adding new lessons, levels, questions, and media without requiring frontend code changes for standard content updates.
NFR-006	Maintainability	The system should separate content data, learning rules, and presentation so that educational content can evolve without unnecessary application changes.
NFR-007	Localization Readiness	The system shall support both Arabic and English content and should be designed to handle right-to-left and left-to-right presentation requirements.
NFR-008	Reliability	Quiz result submission should be designed to minimize accidental loss of completed answers or scores.
NFR-009	Security	Management and instructor functionality shall be protected according to the final authentication and authorization design.
NFR-010	Privacy Minimization	The initial design shall minimize collection of personal student data because individual student profiling is not required by the business scope.

11.1 Natural Voice Requirement

Because audio is part of the educational experience, voice assets shall meet the intended child-oriented quality bar. The final acceptance standard should cover clarity, pronunciation correctness, pacing, tone, and avoidance of an unpleasant robotic effect.

12. Data and Domain Model (Logical View)

The following logical entities describe the minimum business concepts implied by the current requirements. This is a domain-level view, not a database schema.

Entity	Purpose
Class	Represents the shared learning group. Stores class identity, instructor assignment, active level, and academic-year context.
Instructor	Represents a teacher who may be assigned to one or more classes.
Academic Subject	Represents Arabic or English within the Academic Island.
Learning Level	Represents a curriculum stage and its instructional model.
Learning Content	Represents the instructional content elements for a level, including text, images, audio, video, and interactive components.
Quiz	Represents an assessment associated with a learning level.
Question	Represents an individual assessment item with a supported interaction type and answer definition.
Quiz Attempt	Represents a learner's completed interaction with a quiz. It is used for scoring and class aggregation without requiring a persistent student profile.

Entity	Purpose
Media Asset	Represents an image, audio file, video, or other media resource used by learning content.
Academic Year	Represents the period in which class progress is tracked.

12.1 Logical Relationship Summary

- An Instructor may teach multiple Classes.
- A Class has one assigned Instructor in the initial model.
- A Class has one active Learning Level at a time.
- A Learning Level belongs to an Academic Subject.
- A Learning Level contains Learning Content and Quizzes.
- A Quiz contains Questions.
- A Quiz Attempt is associated with a Class and a Quiz.
- An Academic Year provides the time context for Class progress.

13. Business Rules

Rule	Statement
BR-001	Every class shall have an assigned instructor.
BR-002	An instructor may be assigned to multiple classes.
BR-003	Students in the same class follow the same active learning level.
BR-004	Students do not need individual authenticated accounts for the initial product.
BR-005	Class performance is the primary reporting unit.
BR-006	Individual student performance history is not a required business capability for the initial release.
BR-007	The Academic Island supports Arabic and English.
BR-008	A learning level is defined by its instructional model and learning objective, not by a single quiz type.
BR-009	The supported quiz interaction types are Drag and Drop, Matching, Typing, and Choose.
BR-010	Learners may repeat quizzes.
BR-011	Initial question scoring is based on correctness.
BR-012	Class progress is measured by the level reached within the predefined academic curriculum.
BR-013	Learning content must conform to a predefined structure so the child-facing frontend can render it consistently.
BR-014	Computer Island is not part of the current detailed scope.

14. Security and Privacy

The initial design intentionally reduces personal-data collection for students. Because students do not need individual accounts, the system should avoid collecting or persisting personal student information unless it is explicitly required later.

- Management and instructor access shall use an appropriate authentication mechanism for staff users.
- Authorization shall restrict staff capabilities according to the final role-permission matrix.
- Student-facing access shall not expose management functions.

- Class-level results shall be protected from unauthorized administrative access.
- The system should minimize storage of personal data in the learner-facing experience.

15. Integration and Technical Considerations

15.1 Frontend / Content Separation

The frontend should render content based on a structured model supplied by the application/backend layer. The content model should identify the intended component type and its data so that child-friendly presentation can be handled by reusable UI components.

15.2 Extensibility

- The architecture should allow additional subjects and levels to be added without restructuring the entire application.
- Additional quiz interaction types may be introduced later.
- The scoring service should be designed so that the initial simple scoring algorithm can be replaced or extended later.
- The island/world structure should allow future educational domains such as Computer and Music to be added.

15.3 Right-to-Left and Left-to-Right

Because Arabic and English are both supported, the application shall be designed to support both RTL and LTR presentation where appropriate, including text alignment, reading order, layout direction, and child-facing interactions.

16. Out of Scope

- Detailed Computer Island functionality; to be specified in a future phase.
- Individual student authentication, accounts, and persistent profiles.
- Individual student performance dashboards.
- A detailed staff permission matrix beyond the confirmed user roles.
- Advanced scoring algorithms beyond the initial correctness-based model.
- Final UI pixel specifications, visual assets, animation specifications, and branding guidelines.
- External integrations unless separately specified.
- Parent-facing features unless separately specified.
- Payments, subscriptions, or e-commerce functionality unless separately specified.

17. Assumptions and Constraints

ID	Assumption / Constraint
A-001	The Academic Island is the primary delivery scope for the current phase.
A-002	Arabic and English are the only academic subjects required in the current scope.
A-003	All students in a class follow one shared active level at a

ID	Assumption / Constraint
	time.
A-004	Students access learning through class selection rather than individual accounts.
A-005	Class performance is more important to the academy than individual student analytics.
A-006	Quiz scoring begins with a simple correct/incorrect model.
A-007	Repeated quiz attempts are allowed.
A-008	Educational content will be maintained dynamically within a predefined structural model.
A-009	The exact person responsible for content administration is not yet confirmed.
A-010	The exact aggregation rule for repeated quiz attempts is not yet finalized.

18. Open Questions / Items Requiring Confirmation

ID	Topic	Question
OQ-001	Class Entry	Should students type the class name exactly, select from a list, or use a short class code?
OQ-002	Student Session	Should the selected class remain attached to the learner session until the learner exits, refreshes, or explicitly changes class?
OQ-003	Avatar Persistence	Should the learner-selected avatar be retained only in the current session or persist for a longer period?
OQ-004	Display Name	Should the learner display name be stored at all, even temporarily?
OQ-005	Quiz Score Display	Should the learner see a percentage, stars, points, or another representation after completing a quiz?
OQ-006	Repeated Attempts	When a learner retakes a quiz, should class aggregation use the latest attempt, best attempt, all attempts, or another rule?
OQ-007	Level Completion	What exact condition marks a level as completed: instructor action, minimum score, completion of all quizzes, or another rule?
OQ-008	Instructor Actions	Beyond setting the active level, what class-management actions should the instructor be allowed to perform?
OQ-009	Manager Actions	Is the Manager / Head dashboard read-only, or may the Manager / Head also create/edit classes, instructors, or curriculum data?
OQ-010	Content Ownership	Which role is responsible for creating and updating learning content?
OQ-011	Content Workflow	Is content publishing/approval required, or are changes immediately visible after saving?
OQ-012	Arabic Curriculum	What is the finalized sequence and number of Arabic levels for the first release?

ID	Topic	Question
OQ-013	English Curriculum	What is the finalized sequence and number of English levels for the first release?
OQ-014	Avatar Meaning	For educational explanations, what exact rule determines which avatar/character represents a letter, word, or sentence concept?
OQ-015	Media Standards	What are the required recording, video resolution, file-size, and delivery standards for educational media?
OQ-016	Dashboard Detail	Which additional management metrics, filters, and time ranges are required beyond class score and current level?

19. Acceptance Criteria Summary

The initial product scope can be considered functionally aligned with this specification when the following outcomes are demonstrable:

- A child can enter the application, identify the intended class, and access the class's active learning level without a personal login.
- The Academic Island provides Arabic and English learning content.
- Each level presents its own instructional model and can include text, images, audio, video, and visual/avatar-based explanations as configured.
- Arabic and English learning content can include natural, age-appropriate pronunciation audio.
- The quiz engine supports Drag and Drop, Matching, Typing, and Choose question types.
- Quiz answers can be evaluated as correct or incorrect and contribute to the initial simple score.
- A learner can retake a quiz.
- All learners within the same class follow the same active level.
- An instructor can teach multiple classes and can set the active level for a class.
- The system can calculate and present an aggregate class score.
- The system can show the current curriculum level reached by each class.
- The Manager / Head dashboard can present the required class-level score and progress metrics.
- Educational content can be changed dynamically without hard-coding every lesson into the frontend.
- The structured content model can support child-friendly frontend rendering.

Document status: This specification captures the currently agreed initial product direction. Items listed as open questions should be resolved before they are treated as final implementation requirements.